OpenShift Backup and Restore

Currently for Backups on the OpenShift Platform, we run the OpenShift APIs for Data Protection (OADP) Operator. OADP is based on Velero, and provides the standard Velero APIs.

Overview	
Scope	OpenShift Container Platform GA
Status	DEV
Audience	Container Platforms Application Developers Storage and Backup teams
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TODO	Custom SNOW form that covers: • Customized Backup Schedules • One-time manual backups • Add resources to backups (e.g. PODs) • One-time scheduled restore

Setup

To support backups and restores in the environment, we have integrated OADP with Netapp OnCommand for an external (to the cluster) S3-compatible storage backend.

We create a `DataProtectionApplication` that defines a Storage Location named `default` in every cluster. Storage buckets are isolated per environment to avoid mixing data.

Application teams will not have direct access to our backend storage systems. Instead, applications will be backed up using a `Schedule` Custom Resource owned by the Platform Team.

> This solution is ***not*** intended for large-data workloads. If you have are planning to, or consistently have, 500GB or more stored in one individual volume, please reach out to the Container Platforms team for consultation. The process outlined below is intended for smaller data workloads, and has not been tested for large-data use-cases.

Backup

Schedules

Our standard backup schedule is to execute all backups at 2am EST. If you need to request a custom backup schedule, please file (TODO) SNOW form.

Manual Backups

Backups must be executed either by an existing schedule or cluster-privileged user.

To request a manual backup consistent with your change schedule, please file (TODO) SNOW form. Our Tier 2 resources will be able to take a manual backup.

> ⚠ Please allow us a 24-hour acceptance window for these requests. To ensure our support team has the proper data, this workflow is **not** to be executed as a CTASK or during an Incident.

Details

Our standard backup procedure is to whitelist the objects being backed up. These can be configured, but by default we only backup:

- PersistentVolumeClaims
- Persistent Volume Data

If you need other resources included in your backup, please file (TODO) SNOW form.

Persistent Data

Assuming you are using a supported storage class, the Backup process is as follows:

1. Create a VolumeSnapshot using the associated VolumeSnapshotClass
    a. This creates a point-in-time backup at the time of the backup creation
2. Restore a PVC from that VolumeSnapshot into the Backup Controller Namespace
3. Create an exposure pod in the Backup Controller Namespace, mounting the PVC created from step 2
4. The node-agent (running on the infra nodes) will then connect to the exposure pod and upload the volume data into s3
    a. We have chosen the `kopia` backup library as the standard. This is much more performant than restic in the "multi-file" use case, but potentially slower in the "fewer large-file" use case
    b. This operation is incremental. Performing a full backup is only done on the first backup.

> ⛔ If any specific resource needs to be excluded from backups, it requires a label `velero.io/exclude-from-backup=true`

Backup Hooks

To support specific workloads, OADP supports custom backup hooks to run custom commands either before or after the backup is complete.

We do not have any components that require this particular use-case within the Platform team. Please see the Upstream Documentation for reference.

To request the addition of a custom backup hook for your workload, please file (TODO) SNOW form and request that your `POD` objects are backed up. Once this is done, we recommend that you add your backup hooks as annotations (see documentation link above).

If you need assistance configuring annotations for backup hooks, please file a Container Platforms - Off Catalog Request and include the following data:

> *We need assistance configuring backup hooks for (Pod/Deployment name) that is using (technology, e.g. "mongodb") on OpenShift.*
> *Namespace: <namespace>*
> *Contact: <email distro or individual work email>*

We prefer to tackle these types of requests in your Dev or QA environment, such that they can be included in your deployment automation for higher environments.

Common Resources to Exclude

As cluster administrators, it is important that we exclude the following resources from all backups

- clusterserviceversions.operators.coreos.com
- endpoints
- endpointslices
- installplans
- internalhubcomponents.operator.open-cluster-management.io
- leases
- datauploads.velero.io
- datadownloads.velero.io

Restores

A restore by default will restore every object that is included in a backup. If an object already exists in the destination, a restore will not touch that resource. For data, it is *critical* that we remove any PVCs in the destination (and the related PV if necessary) before initiating a restore.

In the event of an Incident that requires a restore, please engage Container Platforms via the Major Incident Management process.

If you need scheduled support, such as to test your DR strategy inclusive of your backed-up objects and data, please file (TODO) SNOW form.

Sneak-peak on implementation - https://azdos-prd.fcpd.fcbint.net/AzdoProdDefaultCollection/ContainerPlatforms/_git/gitops-cluster-com.svbank.uat.pde-sbx/pullrequest/34360

This creates:

- DataprotectionApplication
  - NodeAgent daemonset that schedules on workers and infra
  - Velero that deploys on infra
- externalSecret for s3 credentials
- etcdBackup required items
  - PVC
  - Cronjob
  - serviceaccount
  - Velero Schedule
- Modifies Argo to ignore entirely objects used by velero to manage the backup
  - DataUploads
  - DataDownloads
  - Restores
  - VolumeSnapshots
  - VolumeSnapshotContents

Backups are valid and completing properly.